

The empirical recurrence for OEIS A210049, proved by transfer matrix

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Abstract

OEIS A210049 counts arrays of width 6 whose every 2×2 subblock holds a prescribed number of DISTINCT values, which use ALL of the 3 available letters, and which are counted up to renaming. The entry carries an empirical recurrence of order 85 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The condition counts distinct values and so survives renaming, which lets the count of patterns be recovered from counts over fixed alphabets by inverting a triangular system of falling factorials; the words “containing all values” change which row of that inverse is used and nothing else. Each fixed-alphabet count is a walk count on $S = 794$ states in all.

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1 The conjecture

OEIS A210049 is “Number of $(n+1) \times 6$ arrays containing all values 0..2 with every 2×2 subblock having two distinct values, and new values 0..2 introduced in row major order.”. It has offset 1 and begins

2695, 114237, 5169813, 240224641, 11175987745, 518401340537, 23937024685863, 1100584205159383, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n) = 95*a(n-1) - 1900*a(n-2) - 56187*a(n-3) + 1896826*a(n-4) + 10639671*a(n-5) - 701967562*a(n-6) + 182025504*a(n-7) + 145227173547*a(n-8) - 419177692530*a(n-9) - 19353743252952*a(n-10) + 85433587963187*a(n-11) + 1789310766121030*a(n-12) - 9814164235308092*a(n-13) - 120060982729819483*a(n-14) + 765035170825593245*a(n-15) + 6017016207355416460*a(n-16) - 43541812515518188378*a(n-17) - 229165729445431519990*a(n-18) + 1883135755161684947180*a(n-19) + 6681559617272514956311*a(n-20) - 63470576622786136682427*a(n-21) - 148387908315962080864347*a(n-22) + 1696040408430945413751419*a(n-23) + 24419652218689752156*a(n-24) - 36367868110504859946108233*a(n-25) - 27123365069860973680044986*a(n-26) + 631130053365999735*a(n-27) + 119980272688573408489016056*a(n-28) - 8915897918476195422100426614*a(n-29) + 2412340555420151638480078658*a(n-30) + 102896759828837108198775563215*a(n-31) - 68644270700583096858101003448*a(n-32) - 971547304253211509401889501357*a(n-33) + 995012323475586180840182952483*a(n-34) + 7499405881793341784125165632547*a(n-35) - 10232435340791417316913030684426*a(n-36) - 47161621488766896200634807982479*a(n-37) + 80820197825843514088408850293447*a(n-38) + 239855811396646802214363695180830*a(n-39) - 505732140134030094288174473059546*a(n-40) - 972742523593329235306216393345880*a(n-41) + 2543169520372173508071434849734048*a(n-42) + 3058611451317489983173515035561168*a(n-43) - 10342968088581061157016175191676640*a(n-44) - 6977459283642089223709813906160960*a(n-45) + 34073152101317101518285115092132992*a(n-46) + 9114754028315599477362585372463872*a(n-$

4 Each L_i is a walk count

A 2×2 window lies in 2 consecutive array rows, so the window of the last 1 rows is a state and a step appends a row, settling every subblock it completes. Assembling the pieces for $i = 1, \dots, 3$ as a disjoint union, with ι carrying the weights above cleared of denominators and $\tau = \mathbf{1}$,

$$a(n) = \iota^\top M^j \tau, \quad S = 794.$$

The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C -finite.

5 The proof

Lemma 1. *Let $q(t) = t^{85} - \sum_i c_i t^{85-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j , and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+85} - \sum_i c_i M^{j+85-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 2.

$$a(n) = 95a(n-1) - 1900a(n-2) - 56187a(n-3) + 1896826a(n-4) + 10639671a(n-5) - 701967562a(n-6) + 1820$$

for every $n > 85$.

Proof. The vector $w = q(M) \tau$ was formed by 85 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly; those integers vanish from the index corresponding to $n = 85 + 1$ onwards and the run continued until the working vector was identically zero. The rational weights are carried as one common denominator throughout, so no division is performed until the final term. $\square$

6 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 13 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: enumerate every array over $\{0, \dots, 2\}$, keep those that are canonical, that use every letter, and whose every subblock passes, and count them. Neither the transfer matrix nor the falling-factorial inversion is involved, and the published terms came back — including the initial zeros, which are a sharp test of the “containing all values” clause: a small array cannot show 3 letters at all.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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- [3] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.